Hariom Mewada BPS2 Weekly Report

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1. Feb 02 2026 - Feb 07 2026

1.1 To Do

1. Create plan for BPS2.
2. Read some papers related to digital well being, try to find out gapes.
3. Modify the app according to the future plan: Commertial tool or just a research prototype.
4. Try to have some behavior change result from the app.
5. If the result are promising then try some conferences to publish a paper on this work.

1.2 Proposed Changes to the existing system

1. Deploy the existing app on play store.
2. Refine the UI elements to correctly reflect the underlying psychology.
3. Improve app's accuracy and performance as part of RA work.
4. Make the system more secure.

1.3 Summary: Navigating Digital well-Being in Digital era: A scoping review of digital-wellbeing [?]

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1. Most people across generations perceive frequent internet use as normal and acceptable.
2. This study is a scoping review to identify, analyze, and synthesize existing definitions, measurements tools, and intervention strategies related to Digital well-Being(DWB).
3. Research emerging from **Asia** is scarce
4. Existing Definition coverage on three perspectives:
   (a) DWB as emerging concept of well being influenced by technology use.
   (b) As a perceived impact of digital engagement on individual well-being.
   (c) As a balance use strategy within digital environment.
5. Three overarching themes were synthesized – behavioral intention to use technology, eudaimonic well being, and socio culture contextual factor.
6. Measurement tool for DWB remains limited; with high five scales identified but none offering an integrated framework, interventions, particularly that emphasize on digital disconnection, reveal mixed result about their effectiveness.

7. Collectively these findings highlights ambiguities, limited measurement development, and inconsistent intervention outcome.

8. This review advances theoritical clarity by framing DWB as a multi dimensional construct that bridges pyschological and socio technological domain.

9. It also provide practical guidance for developing culturally adaptive interventions and policy frameworks that promote balanced and meaningful digital engagement.

1.4 Detaied Summary

1. According to **Data Reportal 2024** [**?**], the number of internet user increased by **3.2 %** from previous year, reaching **5.56 billion** user in **2024**.

2. This figure represents **67.9 %** of the global population, although reporting delay suggest that the actual penetration rate is even higher.

3. The avg duration of daily internet usage has grown streadily, reaching **6h 38 min** .

4. Internet engagement spans all age groups, though usage tends to decline over age.

5. Digital devices now enable continous interaction anytime and anywhere, However constant connectivity often generate discomfort or a sense of loss when disconnected – a phenomenon of called technological unconsciousness.

6. (Hoehe and Thibaut 2020)[**?**] emphasized that individuals must be intentional, creative, and mindful in their technology use to enhance relationships and producitivy; otherwise, technology risks eroding rather than supporting.

7. As (Hoehe and Thibaut 2020) [**?**] noted, technology itself is value neutral – it can be beneficial or harmful depending on how it is applied.

8. When technology intented to improve life is used without restrain than it can ultimately reduce quality of life.

9. Since quality of life is closed tied to well-being [**?**] [**?**] scholars have begun to recognize the need of a new construct – **digital well being** – to describe well being in the digital age. Yet how individual conceptualize and experience digital wellbeing(DWB) remains insufficiently understood [**?**].

10. The emergence of the term **digital - well being** reflects a broader historical evolution in how societies interpret the relationship between technology and quality of life.

11. Early scholarship on intrernet addiction [**?**] technostress [**?**] laid the conceptual groundwork by examining the psychological costs of rapid oraganization.

12. This study highlighted the paradox of technological progress – it's simulatenous capacity to enhance connectivity and generate new form of strain.

13. Building upon this foundation the notion of Digital well being emerged in in late [**?**] as scholars and industry leaders sought to redefine a healthy relationship.

14. Major corporate initiative further popularized the concept . Google's **Digital well-being** and Apple's **Screen Time** programs, for instance framed digital balance as design priority, emphasizing user awareness and self regulation.

15. This industry efforts marked a pivotal shift from merely preventing overuse to actively designing for well-being, signaling a convergence between technology ethics, public health, and user centric innovation.

16. Thus DWB represents both academic and applied response to evolving socio-technical landscape – bridging earlier studies on technostress and contemporary frameworks that emphasize agency, meaningfulness, and design responsibility.

17. Ghai et al. (2023) [**?**] examined global research trends using biblio-metric approach,revealing that DWB began to emerge as major topic in 2016 and has grown significantly since 2019.

18. This growth correlates with global digital adoption during and after the pendamic, advances in technology, increased affordability of devices, and expanding digital infrastruce enabeling broader connectivity.

19. Ghai et al. continue to characterize DWB research into three categories :

    (a) Studies exploring its relationship with smarphones, social network, wellness, overuse and disconnection, and digital ethics.

    (b) Studies investigating how digital use related to individual well-being, and

    (c) Research focusing on social media role on shaping DWB.

20. The overlaping use of related terms – such as digital wellness, digital minimalism [**?**], digital health [**?**], and digital disconnection [**?**] has blurred conceptual boundary.

21. Oste (2020) viewed DWB as recognizing the digital self – a concious relationship with technology.

22. Abeele (2021 a) described it as the optimal balance between being connected and disconnected, influenced by personal, device-related, and situaltional factors.

23. Gui et al. (2017) expanded the factors to include community dimensions, suggesting that collective norms, values, and expactations also determine DWB.

24. Despite those contribution, a comprehesive and widely accepted definition remains lacking – particulary one that explicitely link DWB to convetional well-being concept.

25. **Google** defines digital well-being as a pleasant state that arrive when people use technology to achieve their goal.

26. Through it's dedicated website, Google promotes awareness of how technology should empower, rather than distract user from meaningful goal.

27. Similarly, major technology firms now allow users to personalize digital environment [**?**], Features such as algorithmic filtering, personalized recommedation, and adative interfaces are designed to enhance user satisfaction.

28. However, such personlization typically targets hedonic outcomes - maximizing pleasure and minimizing discomfort rather than eudaimonic outcomes which involves meaning and self-realization.

29. True digital well being thus engagement with purposeful, growth-oriented, experience [**?**]

30. Another source of confusion arises from measuring DWB through build-in through built in application that monitor screeentime or limit usage [**?**, **?**, **?**, **?**]

31. This tools lacks standarized critieria for determinig optimal digital activity levels, as impact of screen time across individuals.

32. More over imperical studies reveals inconsistent effect of this tools.

33. Ironically, user often rely on their device to help them disconnect – adjusting app settings or selectively disabling distractions.

34. Additionally the use of term **Digital well-being** is completely vague, as many exlcude psychological and emotional dimesnions essesntial to accessing wellbeing.

35. Consequently this tools serve as self-monitoring system whose long term impacts remains uncertain.

36. Similarly, most major technology firm allows user to personalize digital environment. Features such as algorithmic filtering, personlized recommendation, and adaptive interface are designed to enhance user satisfaction.

37. However, such personalization target hadonic outcome – **maximizing pleasure** and **minimizing discomfort**, rather than **eudaimonic** outcomes, which involve **meaning** and **self-realization**.

38. Another confusion arises from measuring **BWB** throguh built-in applications that monitor **scren time** or **limit usage**.

39. This tool **lack standardized criteria** for determining optimal digital activity level, as the **impact** of **screen time** varies across **Individual**.

40. More-over empirical studies reveal inconsistent effects of such tools on well-being.

1.5 Positive points

1. This study aims to extract, organize, and synthesize the current state of knowledge about digital well-being across populations and context. This review sought to identify **Definition**, **Measuremnts**,**Interventions**.

2. This study provide clear conceptual or theoritical foundations for what digital well being actually means.

3. They have used JBI fraamework for scoping review, Unlike systematic reviews, scoping reviews do not require much critical appraisal or meta-analyis which aligns with the exploratory nature of the study.

2. Feb 08 2026 - Feb 15 2026

No. of Hours : 10

2.1 List of task completed

1. Coordinated with sir for the App deployment part.
2. Sent him signed APK to deploy a new app saparate from intentor app.
3. Created privacy policy for JoDo App: link
4. Recorded JoDo's feature demonstration video on Teblet using Emulator(Required on play store).
5. Sent him ScreenShot for various App Interfaces from both Android and Teblet Devices.
6. Created Account deletion form: link
7. Recruited five pilots from my friend club: Aritra, Utkarsh, Hiranmay,Farhan and, mainak.

3. Feb 17 2026 - Feb 22 2026

No of Hours: 15

3.1 To Do

1. Decide How many day's the pilot study to be continued.
2. Decide what questions to ask in Pilot study.
3. Read papers on Digital well being.

3.2 List of completed task

1. Called sir and sent him feature graphics:

Figure 1: Jodo Feature graphics

3.3 Proposed changes to the Lock and Home Screen UI

(a) Make the Lock Screen and Home Screen UI like **Binary Clock**, which can act as friction as well as user can see time too.

2. Sent him all the details like: app description, details of the why we are running forground service and why we are quering all install app packages,

3. my responses **We need to query all packages as our app is launcher app we have customised the app drawer to reflect usage time and launch count of every installed app that's why we need to QUERY_ALL_PACKAGES**

4. Upgraded the app to target and compile sdk 35 accroding to play store requirement and sent him the upgrade .aab file.

5. Upgraded version code and version name for new version deployment on playstore.

6. Fixed errors in the App code that occured due **Compile SDK** upgrade from **34 to 35**, like in switch block we can not non constant identifiers.

3.4 In person feedback received so far

1. Give dropdown to select geneder.

2. Did not understood the usecase of cognitive friction (math single digit addition) on lock screen.

3. UI looks like old school.

4. Bug in unlock count computation logic, currently it is considering lock button as unlock even though it is not actual unlock but a light on.

5. Inconsistent value of launch count on Whats-app popup.

6. User were having trust issue due to asking launcher permission in the beginning, they suggested to make it optional.

7. For one user app was not able to start, device name: **Realme 7, Android 11**

8. For one user device details : **Readme9, Android 13** App was crashing and not able to start.

9. user comment: **aur oneplus m middle s app slide karke baki running apps m move kar paa rahe h**

10. user comment: **Back or home button k zarurat nahi, Like app abhi bhi running h but tujhe msg kar paa raha hu**

11. user comment : **kuch app disable karne ka option dal de na, Otherwise baki appse directly access muskil ho jaa raha h**

12. One user had very trust issue with the app asked so much permission and he ended up uninstalling the app.

13. one user said he is not used such a launcher interface, He want his own phone launcher interface he continued to use app, He also had very much trust issue I had to explain him too much about the data usage and anonymity.

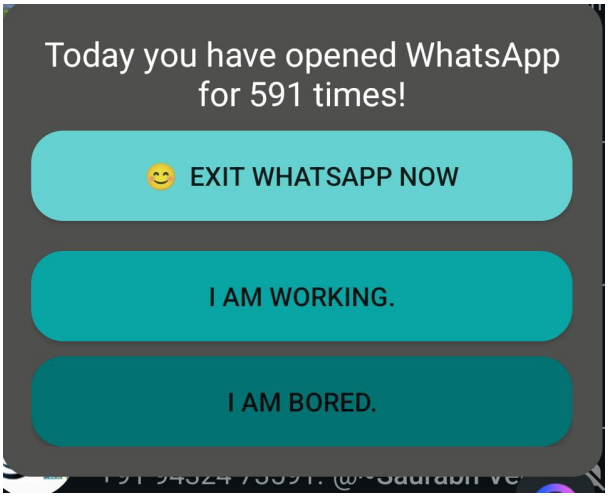

Figure 2: wrong value of launch count

14. User is not able to get the usecase of timer.
15. His response **Muze ye timer ka kuch samajh he nahi ata, kya use hai iska**
16. I told him that this timer is for increasing awareness, his response is: **Nahi, isse kuch farak nahi padta muze, mein switch karke same app open karta hu to phirse counter start hota hai**

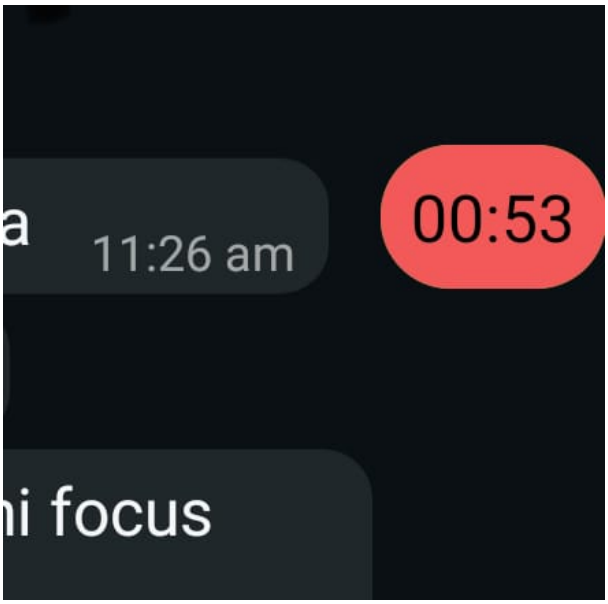

Figure 3: timer running on whatsapp interface

User don't like the UI design for side menu bar. His reponse is: **Ye kuch khas nahi dikh raha**

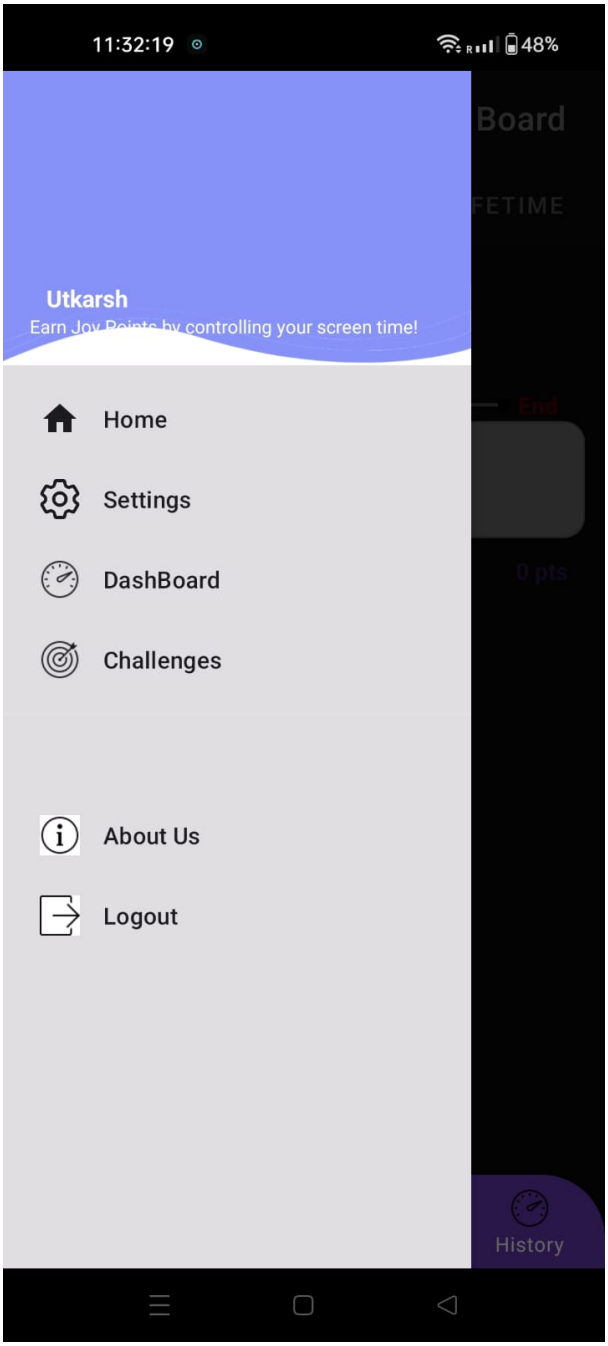

Figure 4: side menu bar interface

3.5 Pilot Study Plan for Jodo App

Objective

The purpose of this pilot study is to evaluate the effectiveness of the Jodo app in improving digital wellbeing and productivity among college students. The study aims to measure whether users experience increased focus time, reduced distractions, improved productivity, and better overall study habits.

3.6 1. Define the Goal

Clearly define measurable goals before starting the study. Example goals:

- Determine whether users increase daily focused study time.
- Measure reduction in screen distractions.
- Evaluate usability and user experience.
- Assess perceived productivity improvement.

Example measurable statement:
Users increase daily focused study time by at least 30 minutes within 7 days of using Jodo.

3.7 2. Select Participants

- Recruit 4-5 college students.
- Include students from different semesters or departments.
- Include both heavy and moderate phone users.

This ensures a diverse and relevant sample group aligned with the app's target audience.

3.8 3. Decide Study Duration

- Minimum duration: 7 days.
- Recommended duration: 14 days.

Digital wellbeing improvements require at least one week of consistent usage to observe meaningful trends.

3.9 4. Pre-Study Survey (Day 0)

Collect baseline data before participants begin using Jodo. Questions may include:

- Average daily study time?
- Average daily screen time?
- Biggest source of distraction?
- Rate productivity (1–10).
- Rate stress level (1–10).

3.10 5. Post-Study Survey (After 7 Days)

Collect feedback and outcome data:

- Did Jodo help you focus? (1–10)
- Did your study time increase?
- Did your screen time reduce?
- Which feature helped the most?
- What was confusing or difficult?
- Would you continue using Jodo? (Yes/No)
- Would you recommend it to others? (Yes/No)

3.11 6. Track In-App Metrics

If available, record:

- Number of focus sessions completed
- Total focus minutes
- App blocking usage
- Daily streak consistency

This provides objective behavioral data in addition to survey responses.

3.12 7. Conduct Short User Interviews

Interview at least 5 participants for deeper insights. Ask:

- When did you feel like stopping the app?
- What feature was most helpful?
- What was frustrating?
- What would you use instead if Jodo did not exist?

Qualitative feedback often reveals usability improvements and hidden pain points.

3.13 8. Data Analysis

Summarize results using a comparison table:

Metric	Before	After
Average Study Time	2 hours	2.7 hours
Average Screen Time	6 hours	5 hours
Productivity Rating	5/10	7/10

Calculate:

- Percentage improvement in focus time
- Dropout or uninstall rate
- Most frequently appreciated feature

3.14 9. Success Criteria

The pilot can be considered successful if:

- 60% or more participants want to continue using Jodo.
- 50% or more report increased focus time.
- Uninstall rate remains below 30%.
- Clear and actionable feature improvements are identified.

3.15 Conclusion

This pilot study will help validate the effectiveness, usability, and engagement level of the Jodo app among college students. The insights gained will guide feature improvements and prepare the app for broader deployment.

4. Feb 23 2026 - Mar 01 2026

No. of Hours: 15

4.1 To Do

1. Collect log files from various pilot devices and analyze it.
2. Create script to do data-analysis on pilot's usage data.
3. Complete post usage interview of the pilots.
4. Prepare final report for pilot study.

4.2 List of task completed

1. Installed app in one more pilot phone(Samsung).
2. Currently 4 pilots are using the app.
3. Three pilots completed 7 days on the app.
4. created github repo (link) for reports.
5. Created substructure for worklog, RA work, running ms report, running ms pres.

4.3 Proposed changes for reward model

New Proposal

1. Like forest app they plant trees we can do water cleaning, pollution control stuff, education support for poor people. Users focus (Pomodoro / deep work sessions).
2. For every X minutes completed:
   (a) Fund water purification
   (b) Support air pollution control
   (c) Sponsor education hours
   (d) Plant trees
   (e) Waste recycling credits
3. Example IIT Bombay competition: "Which department funded more education hours this month?"

4.4 Problem encountered in JoDo

1. App keeps logging out in new devices (pilots).
2. Bugs in celery scheduler, container keeps stopping after throwing exception related to blocking.

4.5 Feedback Recieved about JoDo

1. Pilot3 is also not comfortable with the HomeScreen.
2. Pilot3 removed App as HomeScreen but still keeps using the App.

4.6 Pilot study user feedbacks

1. Change the default timer value if user click I am bored change it 30 minutes, and user click if am working change to 10 minutes.

4.7 New Questions

1. Why Being HomeScreen(Fully minimalist and less visual cues) disliked by users?

5. Feb 26 (last meeting) - Mar 04 2026

No. of Hours: 20

5.1 List of Task Completed

1. Created a python Script to retrieve pilot study data from db and generate graphs and insights over them.
2. Fixed bugs and verified script output.
3. Generated pilot Study data analysis report (link).
4. Collected log file from pilots phone.
5. In one device found Home screen was crashing due to Integer overflow exception.
6. App was crashing on Device BootUp in pilots phone.
7. Prepared new UI design proposal for HomeScreen, Lock Screen and App Drawer.
8. Created Google form (link) for post pilot study feedback, similar to what Omkar used for Intentor user study.
9. Collected response from all the participants.

5.2 Proposed new UI for JoDo

5.2.1 Problem in the current UI

1. It looks like old school.
2. Progress bar has occupied to much space and it is the main cause of so runtime exceptions in the HomeScreen.
3. Current UI not properly reflect out Psychology of **JoDo** as well as **Digital Minimalism**

5.2.2 Proposed UI Prototypes:

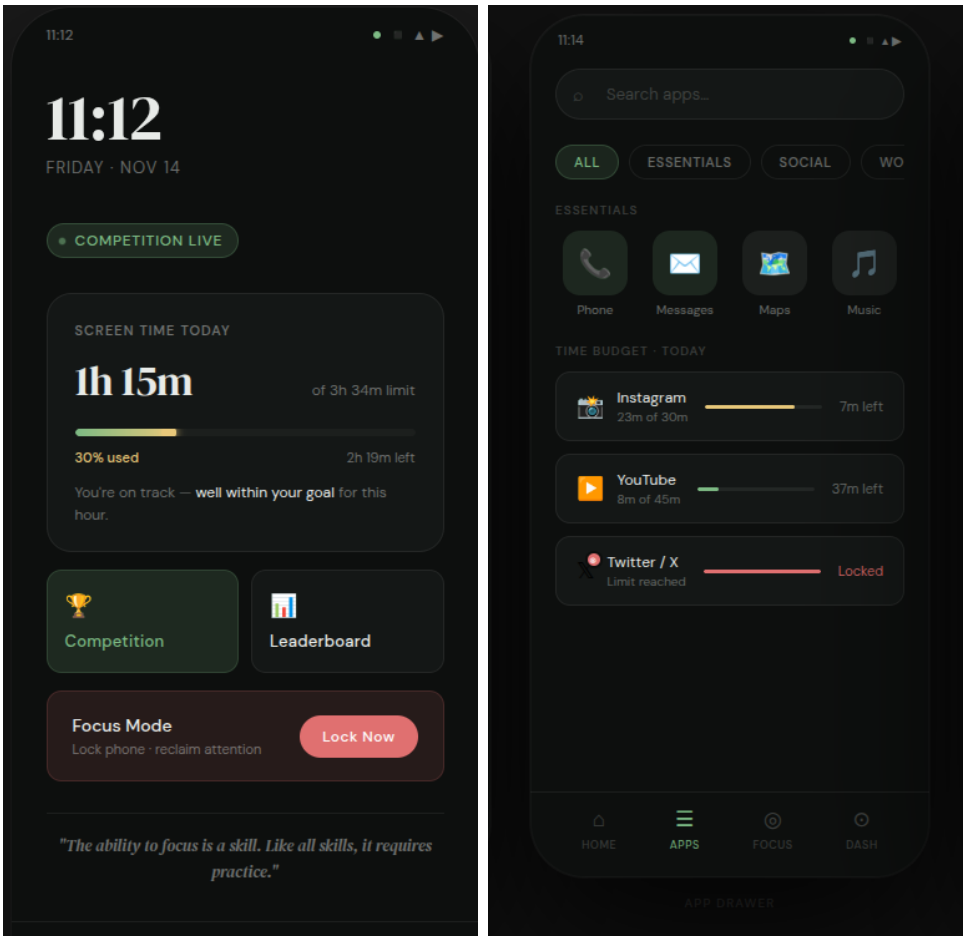

Figure 5: Home-screen(left), App-drawer(right)

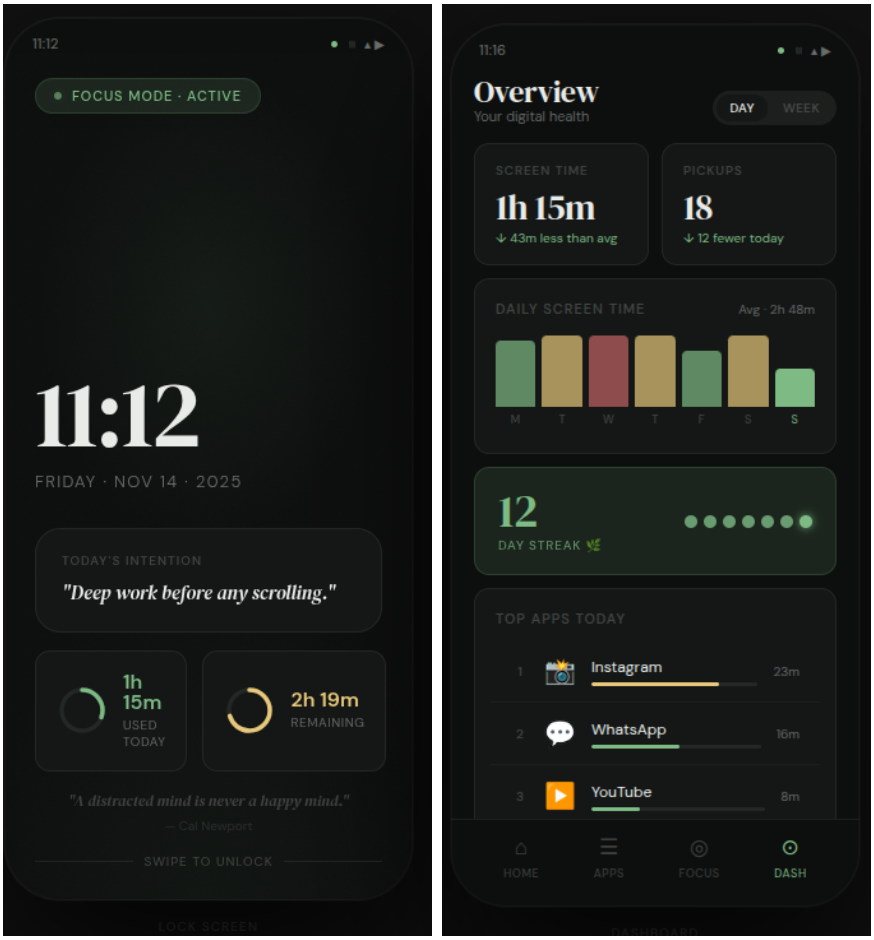

Figure 6: Lock-screen (left), DashBoard (right)

5.3 To Do

1. Create summary for qualitative analysis through the response recieved in google form from all the 3 participants.

5.3.1 JoDo economy similar to Sweat Coin App

1. Create JoDo Economy similar to sweatcoin, they convert steps into real money (but their rewards are virtual).

sweat_coin_app_UI.jpeg